

MedRAG

Evidence-grounded clinical guideline question answering

From scattered clinical evidence to a traceable answer.

MedRAG retrieves relevant passages from indexed guidelines and PubMed abstracts, then uses an LLM to generate a concise answer with evidence, sources, and confidence.

Problem statement

Clinical guidance is extensive, frequently updated, and distributed across long-form documents and research sources. Finding the most relevant passage and explaining it consistently is slow, while an unconstrained language model can produce answers without a reliable evidence trail.

Context

MedRAG applies retrieval-augmented generation (RAG) to clinical guideline Q&A.; A domain-agnostic core handles indexing, retrieval, generation, evaluation, and service orchestration. A MedRAG plugin supplies clinical ingestion rules, metadata, prompts, and collection configuration.

Objectives

- Retrieve clinically relevant evidence from approved indexed sources.
- Generate readable answers that expose evidence and source provenance.
- Provide consistent access through CLI, FastAPI, and Streamlit interfaces.
- Evaluate response quality with automated tests and repeatable golden-dataset evaluations.

01

Grounded answers

Uses retrieved guideline and research context instead of free-form generation.

02

Traceable evidence

Returns evidence snippets, sources, metadata, and a confidence level based on retrieved context.

03

Managed knowledge

Supports PDF upload, deletion, source status, and explicit collection rebuilds.

04

Educational boundary

Returns an educational-use disclaimer and is designed as evidence assistance rather than clinical diagnosis or prescribing.

OPERATING MODEL

How MedRAG works

The system separates knowledge preparation from live question answering. Indexing is explicit and repeatable; query-time work stays focused on retrieval, grounded generation, and structured response delivery.

Stage	Conventional search	MedRAG workflow	System output
Source preparation	Manually locate and scan documents	Parse PDFs and fetch configured PubMed abstracts	Normalized clinical documents
Knowledge organization	Rely on filenames or ad hoc notes	Chunk, enrich metadata, embed, and index in Qdrant	Searchable evidence collection
Question answering	Open documents and synthesize manually	Retrieve top-k context and generate a grounded answer	Answer, evidence, sources, confidence
Quality evaluation	Depends on individual review practice	Run automated tests and DeepEval metrics against the golden dataset	Faithfulness, answer relevancy, and contextual relevancy metrics

Key inputs

- Clinical guideline PDFs placed in or uploaded to the MedRAG guideline source directory.
- Configured PubMed search queries and retrieval limits.
- Runtime credentials for OpenAI answer generation and evaluation.
- Index configuration: embedding model, chunk size, overlap, query mode, and top-k.

Structured output

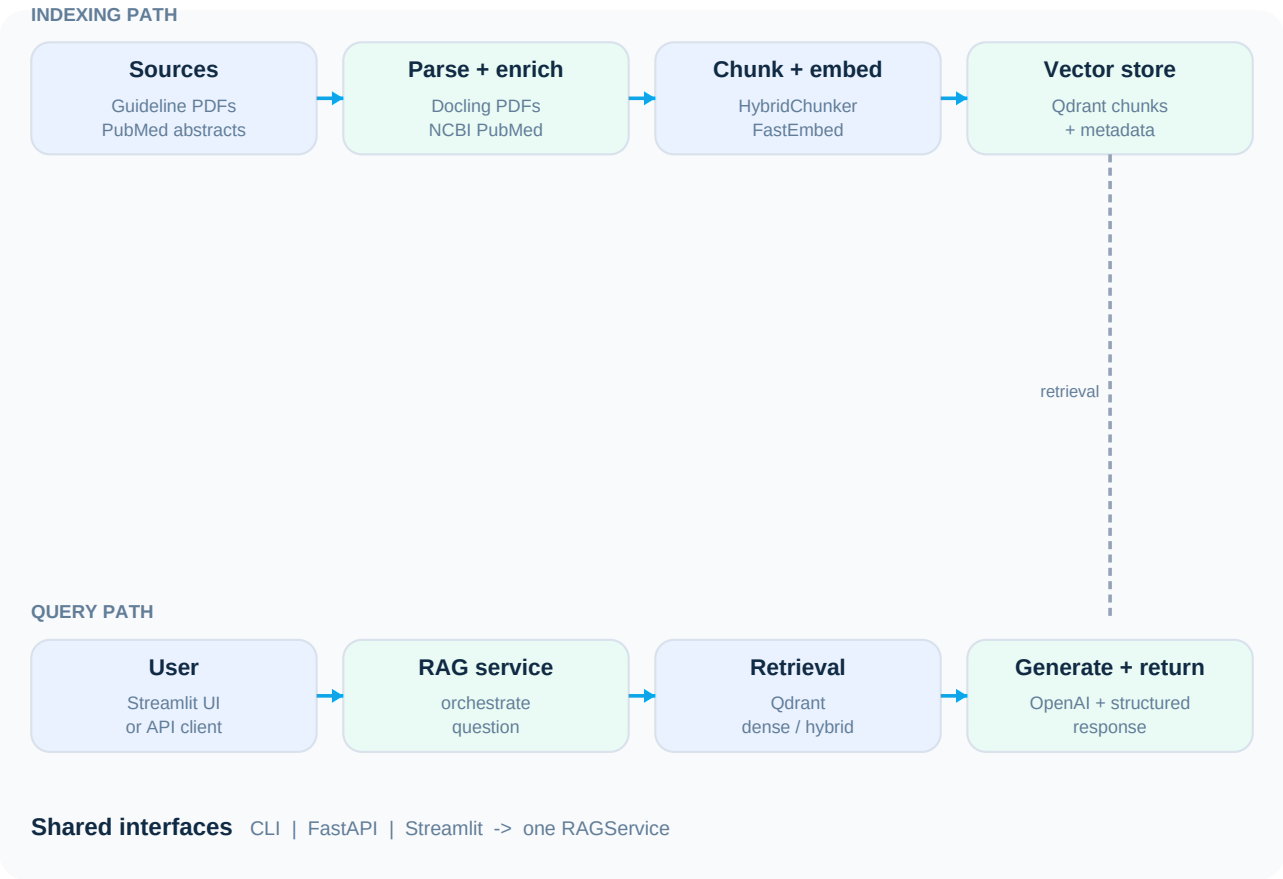
Answer	Evidence	Sources	Confidence
Guideline-level natural-language synthesis	Retrieved supporting snippets	Source file, title, organization, specialty, evidence type	Categorical confidence level based on the amount of retrieved supporting context

No time-saving benchmark is claimed here: the repository defines the automation path, but does not contain a controlled productivity study.

SYSTEM DESIGN

Architecture

One shared RAG service coordinates a plugin-based core. The MedRAG project definition selects its clinical configuration and ingestor; the three interfaces reuse the same retrieval and generation path.



1. Collect and parse

Docling converts clinical guideline PDFs into structured documents for RAG. PubMed abstracts are retrieved through the NCBI E-utilities API. Optional bootstrap documents support demonstrations.
2. Enrich and index

MedRAG labels each document with source organization, specialty, and evidence type. Guideline PDFs are chunked with Docling HybridChunker, while PubMed and bootstrap text use the plain-text fallback. Chunks are embedded locally with BAAI/bge-small-en-v1.5 and written to a named Qdrant collection.
3. Retrieve and generate

A user question becomes a vector search. The selected evidence is inserted into the clinical system prompt, and OpenAI generates a concise educational response.
4. Structure and return

The generated answer is packaged into the MedRAG response schema. The API returns the answer, supporting evidence, source metadata, confidence, and the educational-use disclaimer.

IMPLEMENTATION

Core modules and responsibilities

Layer	Primary modules	Responsibility
Interfaces	src/cli.py src/api/main.py src/ui/app.py	Developer commands, typed HTTP endpoints, and an interactive Streamlit client.
Service	src/core/service.py	Creates a project-specific service, builds or loads the index, and coordinates queries.
RAG core	indexer.py retriever.py generator.py	Chunks and embeds documents, retrieves evidence, builds grounded prompts, invokes generation, and packages structured responses.
Project plugin	projects/medrag/config.py projects/medrag/ingestor.py	Defines clinical prompts, metadata, collection settings, and source ingestion.
Operations	source_manager.py evals.py docker-compose*.yaml	Manages source files, runs quality checks, and packages local or production services.

API surface

Knowledge	GET /sources, POST /sources/upload, DELETE /sources/{filename}, POST /sources/reindex
Question answering	POST /query
Evaluation	POST /evals/medrag/run, GET /evals/medrag/latest
Operations	GET /health

Deployment topology

Docker Compose runs Qdrant, a one-shot indexer, FastAPI, and Streamlit. The indexer waits for Qdrant and prepares the collection; the API starts after indexing; the UI calls only the API. Production assets add Nginx, EC2 bootstrap scripts, CloudFormation infrastructure, and GitHub Actions workflows.

TRUST AND QUALITY

Trust, evaluation, and operational controls

Control	Implemented behavior	Important boundary
Grounded generation	OpenAI receives retrieved MedRAG context together with the clinical prompt.	Grounding quality depends on the indexed sources and retrieval quality.
Educational boundary	Responses include an educational-use disclaimer and expose supporting evidence.	The system does not diagnose, prescribe, or replace clinical judgement.
Evaluation	DeepEval measures faithfulness, answer relevancy, and contextual relevancy against a golden dataset.	Metrics are model- and dataset-dependent and live evaluation requires configured OpenAI API access.
Operational validation	Automated tests, health checks, source status, and Qdrant readiness support system validation.	Production use still requires monitoring and formal clinical, privacy, and security review.

Operational checklist

- Use only sources that the deployment is authorized to process and retain.
- Keep the source collection current, then rebuild the index after source changes.
- Protect API keys and never commit populated environment files.
- Review retrieved evidence and answers for clinical appropriateness before relying on them.
- Run automated tests and the golden-dataset evaluation after prompt, model, embedding, or retrieval changes.
- Monitor API availability, Qdrant availability, response failures, latency, and source freshness in production.

MedRAG should be positioned as an educational evidence assistant. It does not diagnose, prescribe, replace a clinician, or guarantee that indexed guidance is complete or current.

PRODUCT EXPERIENCE

Product walkthrough

The Streamlit application is a thin HTTP client over the FastAPI service. It centralizes knowledge management, question answering, and evaluation results without embedding retrieval logic in the UI.

MedRAG

Clinical evidence assistant

Ask

Sources

Evaluations

Ask a clinical guideline question

Answers are grounded in indexed evidence and are educational only.

What do guidelines say about first-line therapy for hypertension?

ASK

Evidence-grounded answer

Guidelines generally recommend lifestyle changes and selected first-line antihypertensive classes, individualized by comorbidities and clinical context. A qualified clinician should interpret targets and treatment choices.

CONFIDENCE: HIGH

Evidence and sources

WHO_BP.pdf

guideline | cardiology

PubMed abstract

research_abstract | cardiology

Typical user flow

1	Prepare	Upload approved guideline PDFs and inspect PubMed ingestion settings.
2	Index	Rebuild the Qdrant collection so new or removed sources affect retrieval.
3	Ask	Submit a clinical guideline question through the UI or POST /query.
4	Review	Read the answer together with evidence snippets, sources, and confidence.
5	Assess	Run the evaluation suite after meaningful system changes.

Project status and next priorities

- Current scope: RAG core, MedRAG plugin, source management, typed API, Streamlit UI, evaluations, and Docker/AWS deployment assets.
- Next: set product quality thresholds, expand the clinician-reviewed golden dataset, and strengthen production monitoring and failure handling.
- Before sensitive use: add authentication, authorization, audit logging, retention rules, and formal privacy, security, and clinical review.

Prepared from the MedRAG repository implementation and the supplied report's presentation pattern. Repository state reviewed 16 September 2026.